Cascaded $\chi^{(2)}$:$\chi^{(2)} \rightarrow \chi^{(3)}$ in single-resonance $LiNbO_3$ metasurfaces

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The idea, and the metric

Cascaded Kerr. Two $\chi^{(2)}$ steps synthesise an effective Kerr response on the pump; the second harmonic is eliminated adiabatically:

$$\omega_1 + \omega_1 \xrightarrow{\chi^{(2)}} 2\omega_1 \xrightarrow{\chi^{(2)}} \omega_1, \qquad \chi^{(3)}_{\mathrm{casc}}(\Delta\omega) = -\frac{\omega_1 \, |\chi^{(2)}|^2 \, \eta_{\mathrm{geom}}}{6 \, n_2^2 \, (\Delta\omega + i\gamma_2/2)}.$$

Single-resonance (SH-only) case: only the $2\omega_1$ mode is resonant — far easier to fabricate than a two-mode device.

The figure of merit

$$R \equiv \frac{|\chi^{(3)}_{\mathrm{casc}}|}{|\chi^{(3)}_{\mathrm{native}}|} \;\propto\; Q_2 \cdot \eta_{\mathrm{geom}} \;\propto\; Q_2 \cdot (\text{SHG overlap})^2.$$

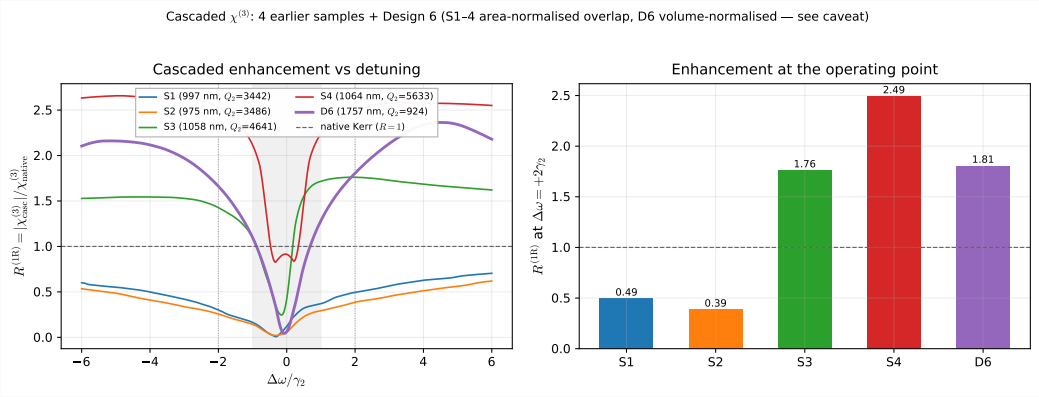
$R > 1$: the cascade beats the material's own Kerr. Operating point: conservative $\Delta\omega = +2\gamma_2$.

Five $LiNbO_3$ candidates

	λ_1	Q_2	SHG overlap	η_{geom}	R	beats native?
S1	997 nm	3442	0.12	0.014	0.49	**no**
S2	975 nm	3486	0.11	0.011	0.39	**no**
S3	1058 nm	4641	0.19	0.037	1.76	**yes**
S4	1064 nm	5633	0.21	0.044	**2.49**	**yes**
D6	**1757 nm**	**924**	**0.43**	**0.18**	**1.81**	**yes**

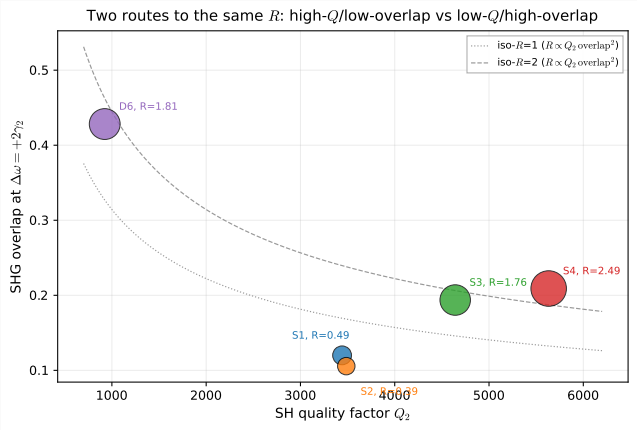
- S1–S4: pure-$LiNbO_3$ pillars, height asymmetry, green SH.
- **D6**: **TiO_2/$LiNbO_3$ hybrid**, etch-angle asymmetry ($0°/90°$), pump at 1.76 μm.

Enhancement over the native Kerr



Only **S3, S4, D6** exceed $R = 1$. Ideal-limit winner: **S4** ($R = 2.49$). The overlap is small at resonance (Fano dip), so the wings are used.

Two routes to the same R

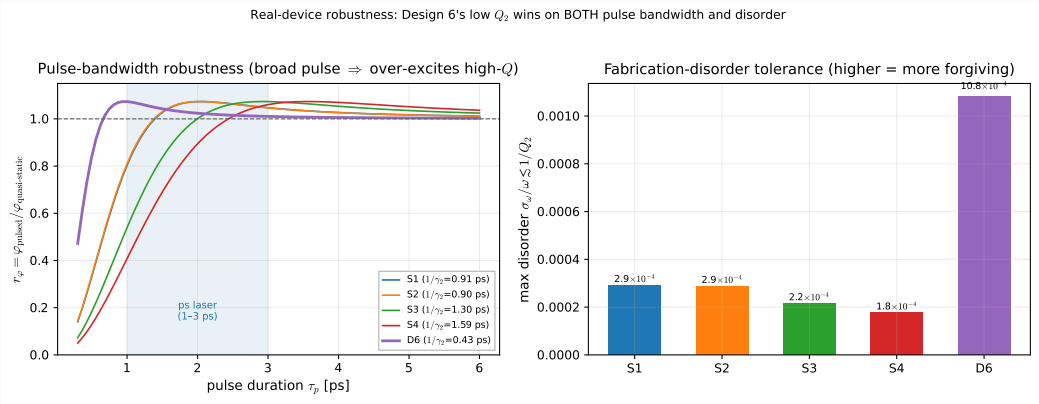


Since $R \propto Q_2\,(\text{overlap})^2$:

- **S3/S4:** high Q_2 ($\sim$5000), modest overlap ($\sim$0.2).
- **D6:** low Q_2 (924), **high overlap** (0.43).

Both reach $R \approx 2$ — but they behave very differently in a *real, pulsed* experiment.

The catch: a real ps pulse is spectrally broad



A τ_p=1–3 ps pulse has bandwidth $\sim 0.5\,\text{THz}/\tau_p$[ps]. For **high-$Q$** designs this *exceeds* the linewidth: up to **22%** of the generated SH lands at the *wrong* detuning sign (cancellation) ⇒ the Kerr phase collapses (r_φ=0.4 at 1 ps for S4). **D6**'s broad linewidth keeps the pulse clean ($r_\varphi \approx 1$) *and* tolerates 5× more fabrication disorder.

Which is best?

Ideal limit

S4, $R = 2.49$.

Highest Q_2, best raw enhancement — *if* the design Q is reached and the pulse is long enough ($\tau_p \gtrsim 3\,\mathrm{ps}$).

Real fabricated device

D6, $R \approx 1.8$.

Nearly the same enhancement, but **robust**: no pulse-bandwidth penalty at 1–3 ps, 5× looser disorder tolerance, low Q reliably fabricated.

High-Q wins on paper; **low-Q/high-overlap wins in the lab, at least theoretically**.

What we measure in the experiment (my subjective – open to discussion)

Closed-aperture Z-scan vs. pump detuning — the cascade's fingerprint.

- Pump always at ω_1; scan the pump wavelength ($\sim$1 nm) so the generated $2\omega_1$ sweeps across the SH resonance, $\Delta\omega : - \to +$.
- **Sign flip:** valley$\to$peak (self-focusing) $\to$ peak$\to$valley (defocusing) as $\Delta\omega$ crosses zero. The cascade is *odd* in $\Delta\omega$; native Kerr is a constant background $\Rightarrow$ isolate the cascade as the antisymmetric part.
- **SHG conversion efficiency** in the same sample calibrates the absolute $\chi^{(3)}$ scale (pins C_{bridge}).

Regime

ps pulses, $\tau_p \sim 1$–3 ps at $\lambda_1 \approx 1.76\,\mu$m (available). D6's SH lifetime $1/\gamma_2 \approx 0.43$ ps $\Rightarrow$ quasi-static, sign flip stays sharp.

Is it achievable in a real device? Preliminary analysis

Goal	Verdict	Why
Beat native Kerr, $R > 1$	yes	$R \approx$ 1.8–2.5 (S3/S4/D6); robust to the $O(1)$ scale uncertainty
Measure the cascade (sign flip)	yes	antisymmetric-in-$\Delta\omega$ signature; cleanest on D6
CW Kerr *device*	no	single-pass phase is μrad even at damage — need a two-mode cavity
Reach the *designed* high Q	risk	disorder caps S4 first ($\sigma_\omega/\omega \lesssim 2 \times 10^{-4}$)

Honest uncertainties. Absolute R is order-of-magnitude (C_{bridge}, native $\chi^{(3)}$ at these wavelengths, the area/volume overlap convention). The *sign*, *ranking*, and Q/η scaling are far more robust than the absolute number.

Bottom line

Recommendation

Run the ps-pulsed **Z-scan sign-flip** measurement on **Design 6** at $\lambda_1 \approx 1.76\,\mu$m, with an SHG-efficiency calibration in the same sample.

- **S4** has the highest ideal R (2.49) but is the most fragile (needs $\tau_p \gtrsim 3$ ps and sub-2×10^{-4} disorder).
- **D6** gives $R \approx 1.8$ (up to $\sim$2.4 in the wings) and is robust on *both* pulse bandwidth and fabrication disorder — the best design to actually *see* the cascaded $\chi^{(3)}$.
- The achievable observable is the **sign flip**.